Hector Lugo III

+1 (956) 999-2401 | hlugo576@gmail.com | linkedin.com/in/hector-lugo-3rd | github.com/Wavy-Hec

Multi-camera video understanding with vision-language models · reinforcement learning for legged robots · neuro-symbolic multi-agent search

Education

The University of Texas Rio Grande Valley Expected Dec. 2026
M.S. in Computer Science (GPA: 3.79/4.00) | B.S. in Computer Science (Aug. 2020 – May 2024)

Experience

AI/ML Research Intern May 2026 – Aug. 2026
Air Force Research Laboratory (AFRL), Rome, NY

- Built and open-sourced the team's evaluation harness for multi-camera video QA: 80,000+ scored runs across 3 benchmarks and 2 vision-language backends (Qwen2.5-VL, InternVL3) on a multi-GPU SLURM cluster.
- Compared centralized fusion of camera feeds against decentralized per-stream agents and benchmarked agentic frame- and clip-selection methods (CLIP/SigLIP retrieval, model-written search queries) against uniform sampling at matched frame budgets; centralized led every comparison, and selection consistently lost to uniform sampling.

Machine Learning Engineer Intern May 2025 – June 2026
Idaho National Laboratory (INL)

- Engineered a ResNet autoencoder to denoise smart-grid sensor signals for anomaly detection.
- Accelerated model training by over 93% (from 30 days to 1–2 days) with signal preprocessing and parallel computing.

Research Assistant Jan. 2023 – Present
Machine Intelligence Lab, UTRGV

- Conduct Vision-Language Model (VLM) and Vision-Language-Action (VLA) research for humanoid locomotion, and study whether a legged robot's spatial memory revises when its environment changes mid-episode.
- Build Isaac Sim environments and train reinforcement learning policies for Unitree Go2 quadruped navigation, including state autoencoders that cut compute cost and speed up training.

Graduate Teaching Assistant Jan. 2024 – Present
The University of Texas Rio Grande Valley (UTRGV)

- Teach CS 1101 (Intro to CS) and Digital Image Processing to 40+ students, handling lectures, labs, and grading.

Data Science Research Intern June 2023 – Aug. 2023
University of California, Riverside

- Applied spectral clustering to U.S. railway accident data, identifying critical spatial safety zones.

Projects

Multi-Agent Neuro-Symbolic Video Search | *Temporal Logic, VLMs* | [code] 2026 – Present

- Designing a multi-agent system that searches asynchronous video streams: per-stream agents evaluate local temporal-logic queries and a central orchestrator fuses them against a global neuro-symbolic specification.
- Extending retrieval into automated response: matches trigger analyst alerts, focused collection, or follow-up queries. Continuing the AFRL collaboration past the internship; manuscript in preparation.

Real-Time Object Detection & Tracking | *YOLOv8, SORT, OpenCV* | [code] 2025

- Built a real-time multi-object detection and tracking pipeline: YOLOv8 detections feed a SORT tracker (Kalman filter + Hungarian matching) that keeps unique per-object IDs across all 80 COCO classes, with FP16 GPU inference.

Personalized Chemotherapy Dosing | *Decision Transformer, Offline RL* June 2024 – July 2024

- Decision Transformer that learns personalized chemotherapy dosing from offline patient data, framed as offline RL.

Publications

- **IEEE ICMLA 2026** — Vision–Language Guided Quadruped Navigation with RL Control (accepted).
- **ICICT 2025** — Personalized Chemotherapy Dosing Through Offline Reinforcement Learning. [paper]
- **ICICT 2025** — Offline Reinforcement Learning for Safe and Effective Smart Grid Control. [paper]
- **ASME JRC 2024** — Spectral Clustering in Railway Crossing Accidents Analysis. [paper]

Technical Skills

Languages: Python, C++, Java, SQL
Frameworks & Tools: PyTorch, Hugging Face, OpenCV, NumPy, Pandas, Git, Docker, Linux, HPC (Slurm)
Robotics & Edge: Isaac Lab, Isaac Sim, MuJoCo, LocoMujoco, ROS, Unitree Go2, NVIDIA Jetson Nano
Concepts: Reinforcement Learning, VLMs/VLAs, Multi-Agent Systems, Multi-Camera Perception, Computer Vision